

Cortical Columns: Redefining AI with the Principles of Human Brain Architecture

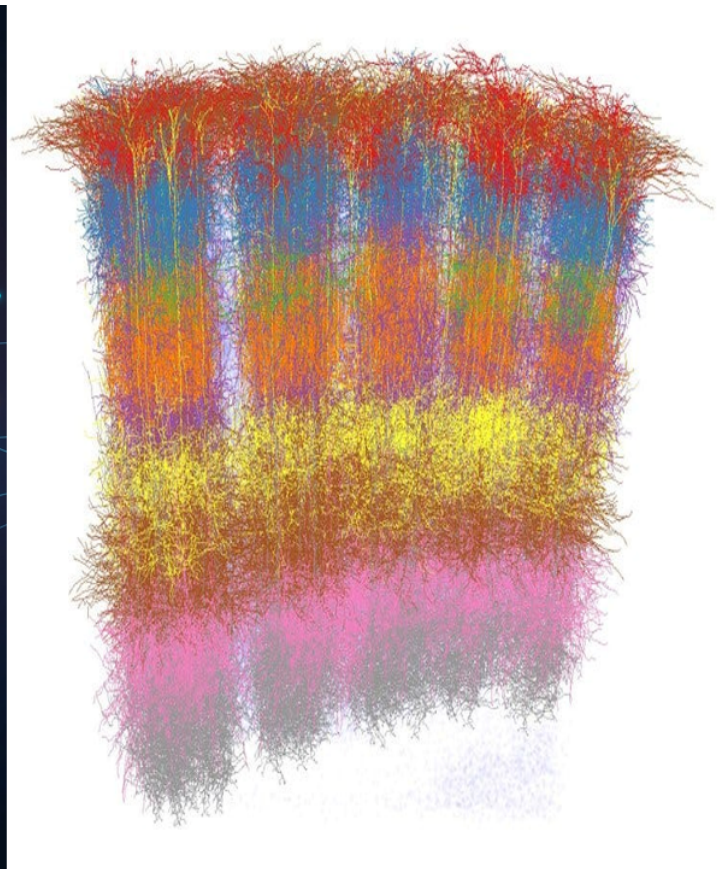
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The image of the cortical columns on the right side was released by Marcel Oberlaender.

Artificial Intelligence (AI) has significantly transformed our lives, becoming an integral part of our daily existence. It has made remarkable strides in various fields, including medicine, customer service, finance, manufacturing, transportation, retail, and education. However, despite these advancements, AI still falls short of the human brain's capabilities — in fact, it's not even as smart as a cockroach.

Consider this: If you place a cockroach in the middle of a forest, it will find its way and survive. In contrast, if we put an advanced robot equipped with an AI-enabled central control system in the same environment, it would likely become lost. Why? The answer lies in the fundamental differences between artificial and biological neural networks.

The Limitations of Current AI

Present-day artificial neural networks are based on decades-old basic concepts of neuroscience. The way artificial neurons are connected doesn't accurately reflect the complex architecture of the human brain. AI systems process data in a step-by-step manner, learning about simple features before progressing to more intricate objects. This approach requires massive static databases for learning and generates a single model of the world. As a result, even slight alterations in the dataset or a single pixel can produce drastically undesired outputs.

The Thousand Brains Project: A New Frontier

Enter the Thousand Brains Project, an open-source initiative working towards developing an entirely new AI framework. The project's founder posits that this framework will be based on the same principles as the human brain, specifically the neocortex structure.

The neocortex, which accounts for about 80 percent of the human brain's mass, is made up of thousands of so-called cortical columns (as depicted in the image). Each column is divided into multiple layers of neurons, with approximately 150,000 cortical columns in the human brain. What's fascinating is that each column carries out its own learning.

Sensorimotor Learning: The Key to Human-like AI?

The cortical columns employ a concept called 'sensorimotor learning'. This approach allows each column to develop a model of the world by perceiving its surroundings through senses and body movements. Due to varying levels of sensory and motor feedback, each cortical column builds its own unique model of the world. This is in stark contrast to deep learning networks, which build a single model of the world and learn from the costly and laborious task of creating and labeling datasets.

The Promise of the Thousand Brains Project

The Thousand Brains Project is working towards building a machine that could function like the brain's neocortex in terms of sensorimotor learning. By mimicking this neuroscience structure in AI, they aim to create systems that can perform sensorimotor tasks in each of the cortical column-like units. These units would communicate using links, similar to the long-range connections seen in our brain structure.

This revolutionary concept holds the promise of achieving artificial general intelligence (AGI) that would closely resemble human intelligence. Such AI systems could potentially carry out intricate tasks with the adaptability and nuance that characterize human cognition.

Looking Ahead

As we stand on the brink of this exciting new frontier in AI development, it's crucial to consider the implications and potential of such technology. While current AI systems have already revolutionized numerous aspects of our lives, the development of more human-like AI could open up unprecedented possibilities across all sectors of society.

However, it's equally important to approach these advancements with caution and ethical consideration. As AI continues to evolve, we must ensure that it serves to enhance human capabilities rather than replace them, and that it is developed and deployed in a manner that benefits all of humanity.

The journey from today's AI to truly human-like artificial intelligence is long and complex, but projects like the Thousand Brains initiative are paving the way for a future where the gap between artificial and human intelligence may finally be bridged.